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Wind Farm Wake-Steering Exploration During Grid Curtailment

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Abstract— **Wind farm wake steering is an active topic in the wind community. Yaw-induced wake steering in wind farms has shown significant increases in total wind farm power output. Unfortunately, sensor uncertainty and model uncertainty often make pure model-driven approaches less effective. Due to the mechanical and financial downsides associated with experimental wake steering, collecting useful data to verify model-based approaches is often viewed as too risky. However, electric grid curtailment periods offer the opportunity to experiment with minimal mechanical and financial risks. A novel data acquisition process is presented that utilizes grid curtailment periods for wake steering experimentation. This method curtails the total wind farm power output while yaw sweeping the upstream turbine to discover the optimal yaw angle for wake-steering. The optimal yaw angle can then be used in regular uncurtailed periods to increase total power output.**

I. INTRODUCTION

As a wind turbine removes energy from the wind, it produces a wake that disperses downstream. This wake can interact with downstream turbines, often increasing loading moments and decreasing electricity production [1], [2]. Yaw-induced wake-steering seeks to mitigate the negative impacts of wakes on downstream turbines [3]. Unfortunately, physical wind farm experimentation to find the optimal upstream turbine yaw angle can be risky both financially and mechanically [3]. Wake-steering models offer a low-risk method for predicting the optimal upstream turbine yaw angle. However, due to model uncertainty and sensor uncertainty, physical execution of yaw-induced wake steering may be less effective when relying solely on model-based approaches [4], [5]. Grid curtailments, controlled reductions in power production that occur if electrical power production in a grid exceeds electricity demanded, offer times for low-risk physical experimentation with minimal inaccuracies. Curtailment periods may occur while turbines are operating in below-rated wind speeds. Wake steering can only improve total farm power output when in below-rated wind speeds. Therefore, below-rated curtailment experimentation will give the most relevant data for uncurtailed wake steering for power optimization. This low-risk, higher-accuracy method, shown in Figure 1, is the topic of this research.

As renewable energy becomes a greater percentage of a grid's overall electricity production, the frequency and magnitude of curtailment periods will likely increase due to the variability of wind and solar resource [6]. Curtailing a wind turbine is typically achieved by pitching a turbine's blades and operating at a non-optimal tip speed ratio. This reduces aerodynamic forces from the wind therefore reducing the amount of energy captured [7].

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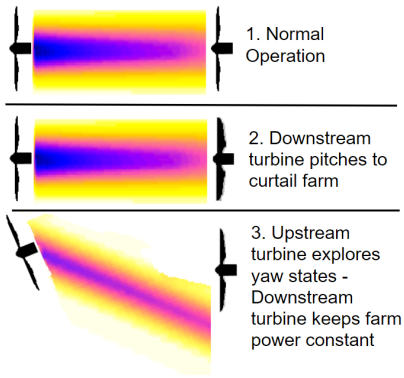


Fig. 1. Exploration technique for garnering useful yaw wake-steering data while curtailing the power output of a two turbine wind farm.

The topics of wind farm wake effects and grid stabilization via wind farm control is an actively researched topic. Coordinated control of horizontal axis wind turbines in a wind farm has shown benefits compared to independent operation in a number of studies [1], [8], [3]. Yaw-induced wake steering is a promising method that can be implemented on existing utility-scale wind farms [9]. Yaw-induced wake steering works by pointing a wind turbine slightly offset from the wind direction. The vector face of the turbine is the approximate force direction applied to the wind, causing the wake to deflect to the side. This can steer the low-energy wind profile in the wake away from downstream turbines thus leaving higher energy wind for capture. Related work has shown the positive potential of wind farms operation during grid curtailment periods [10] [11] [12] [13]. A novel feature of this work relative to the existing literature is how this work seeks to use grid curtailment periods to improve performance during uncurtailed periods.

The remainder of this paper consists of background, approach, results, and conclusions. The background in Section II summarizes models for wind farms and individual wind turbines. It also discusses the benefits and challenges associated with wake steering. The approach in Section III shows how to collect wake steering data while producing a constant, curtailed power output. This same section provides the process for predicting the optimal upstream turbine yaw angle. Section IV presents an example of the proposed methodology and its accuracy while analyzing the data. The conclusions in Section V restates the benefits of this method and then discuss future work.

II. BACKGROUND

A. Individual Turbine Model

The majority of this paper centers around the relation of the turbine operating state in relation to its incoming wind. The power (W) captured from the wind by a single turbine is given as follows [7]:

$$P = \frac{1}{2}\rho A U^3 C_p(\beta,\lambda) \tag{1}$$

where ρ is the air density (set to 1.225 kg/m^3), A is the turbine swept area (m^2), and U is the incoming wind speed perpendicular to the rotor plane. (m/s). The efficiency of the wind turbine is denoted by the power coefficient $C_p(\beta,\lambda)$ which is controlled by changing the turbine's blade pitch angle, β. It also depends on the (non-dimensional) ratio of the tangential blade tip velocity to the incoming wind speed. This tip speed ratio, λ, is given as follows [7]:

$$\lambda = \frac{\Omega R}{U} \tag{2}$$

Ω is the rotor speed (rad/sec) and R is the rotor radius (m).

Figure 2 illustrates the typical dependence of power captured versus incoming wind speed U. Turbine operation is broken down into four different regions depending on the wind speed. The turbine blade pitch, yaw, and generator torque are operated differently in each region.

Region 1 describes low wind speeds that are not significant enough to power the wind generator. In this region, the wind turbine may pitch its blades at a 90 degree angle to feather the blades. This minimizes aerodynamic loads and keeps rotor speed at zero. Region 4 has similar controls objectives to shut down the turbine but for the purpose of load mitigation in high wind speeds.

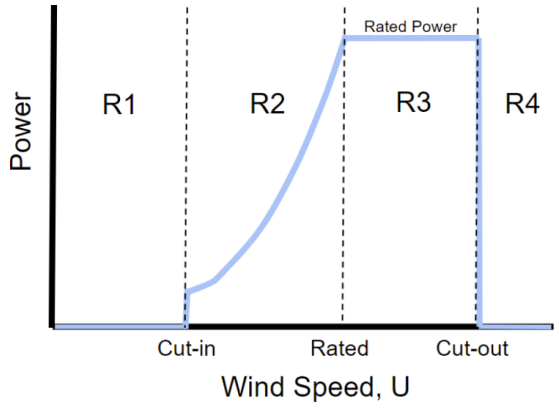


Fig. 2. Wind turbine control regions based on wind speed. The NREL 5-MW turbine has 3 m/s cut-in, 11.4 m/s rated, and 25 m/s cut-out wind speeds. [14]

Region 2 describes moderate wind speeds where the control objective is to capture as much power from the wind as possible. It does this by maximizing its power coefficient $C_p(\beta,\lambda)$. The NREL 5-MW reference turbine (used throughout this paper) keeps its blades at an optimal 0 degree pitch angle, β, in region 2 wind speeds in order to capture as much wind as possible. The generator controls its torque to keep the turbine at the optimal tip speed ratio λ. For the NREL 5-MW reference turbine the optimal λ is 7.55.

Region 3 describes a region of high wind speeds. In this region the turbine limits how much power it captures from the wind, attempting to stabilize at its rated power. In this region the wind turbine varies its blade pitch angle to keep the aerodynamic torque on the rotor constant while generator torque attempts to keep the rotor speed constant. It is of note that any increase in wind speed is not ideal in region 3 due to load increases. Therefore, wake steering is not used to increase power capture when the downstream turbine is in region 3 conditions.

B. Wind Farm Model

The interactions between wind turbines via wakes require additional modeling. If the wake deficit of the upstream turbine "hits" the downstream turbine it reduces wind speed at the downstream turbine and thus decreases available power. Consider a two turbine farm with the free-stream wind speed of U as shown in Figure 3.

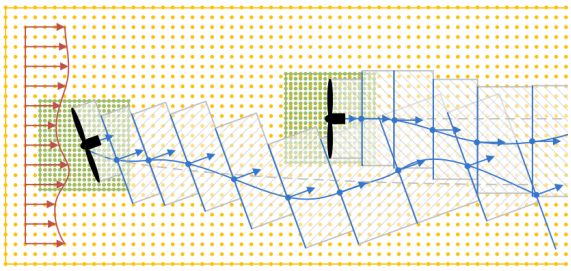


Fig. 3. An upstream turbine yawing offset to the direction of the wind to steer its wake away from a downstream turbine. Varying resolution grids for wake modelling in NREL's FAST.Farm simulator software are shown as dense dots around the turbines and spread-out low-resolution dots in the farm [15].

Let γ_1 denote the yaw offset angle for the upstream turbine relative to the free-stream wind direction. In addition, label the upstream and downstream turbines as 1 and 2, respectively. Thus U_1 and $U_2(\gamma_1)$ are the incoming wind speeds at the upstream and downstream turbines with $U_1 = U_\infty$. It follows from Equation 1 that the captured power is:

$$P_1(\gamma_1) = \frac{1}{2}\rho A U_1^3 C_p(\beta,\lambda,\gamma_1) \tag{3}$$

Note the upstream turbine is assumed to maximize its power capture in both the uncurtailed and curtailed yaw explorations. This is achieved by keeping $\beta = 0$ and $\lambda = 7.55$ as γ_1 varies.

The yaw offset of the upstream turbine also affects the direction of the downstream wake. Thus the incoming wind speed, U_2, and power captured, P_2, by the downstream turbine become a function of upstream turbine yaw angle. Equation 1 is modified to explicitly denote this dependence of the downstream power capture on the upstream turbine yaw:

$$P_2(\gamma_1) = \frac{1}{2}\rho A U_2(\gamma_1)^3 C_p(\beta,\lambda) \tag{4}$$

Since turbine 2 has no yaw offset in these simulations, C_p is not expressed as a function of γ_2. Note this is an assumption that would not apply to real world turbines. The total power of the two turbine wind farm is simply the sum of the power captured from the upstream and downstream turbines:

$$P_{farm}(\gamma_1) = P_1(\gamma_1) + P_2(\gamma_1) \tag{5}$$

To model the wake interactions in this research the National Wind Technology Center's FAST.Farm mid-fidelity wind farm simulator was utilized. FAST.Farm utilizes OpenFAST, an open source wind turbine simulation tool that predicts individual turbine loadings and power output using aero-hydro-servo-elastic engineering models [15]. Blade-Element Momentum (BEM) theory is used to model rotor aerodynamics. FAST.Farm adds wind farm-wide atmospheric boundary layer physics to model wake deficits, advection, deflection, meandering, and merging as a result of wind turbine interaction. Some of the aspects of the Dynamic Wake Meandering (DWM) model make up the wake model of FAST.Farm primarily for wake-deficit evolution and wake meandering [16]. FAST.Farm has been calibrated and validated against SOWFA, a high-fidelity large-eddy simulation tool that also finds wind turbine loading and power outputs [17], [18].

FAST.Farm models winds at varying resolutions: a high resolution for the volumes closest to wind turbines and a lower resolution encompassing the entire wind farm. FAST.Farm intakes a set of wind velocity vectors for each point as shown in Figure 3. These wind velocity vectors describe how the winds in the array would be changing in time if no wind turbines were present in the flow field.

This research focuses on a theoretical two turbine wind farm comprised of two NREL 5-MW wind turbines. An 8 m/s wind was simulated traveling from west to east as shown in Figure 4. The farm was modeled with low and high resolution wind vector input volumes with physical dimensions (length, width, and height), spacing between grid points, and calculation time steps as described in Figure 4.

Here, both the low and high resolution grids follow a basic power law equation with each grid point having roughly the same velocity as the grid points at the same height. Note, this means the incoming wind used here has essentially no turbulence which is not representative of real wind farm winds. The vertical shear power law for wind speed is given as [7]:

$$\frac{u}{u_r} = \left(\frac{z}{z_r}\right)^{\alpha} \tag{6}$$

where u is the wind speed at a given height z. The reference wind speed u_r is given as 8.0 m/s at a hub height z_r of 90m. This yields the hub height wind speeds shown in Figure 4. For this wind profile a standard α of 0.14 represents the increase of wind speed as height increases.

C. Uncurtailed Yaw Exploration

This section describes an uncurtailed yaw exploration method. This provides a baseline for comparison to the proposed method to explore during curtailed operation. For this baseline, the upstream turbine yaw angle γ_1 is held constant for 2000 seconds during an uncurtailed operation. The yaw angle is then increased by 2 degrees and the simulation is repeated with the same incoming wind profile and all other quantities kept constant. This is done for γ_1 from 0 to 28 degrees. Figure 5 shows the power captured by the upstream turbine, downstream turbine, and farm power. The upstream power output, $P_1(\gamma_1)$, varies with the yaw angle as $\cos^3(\gamma_1)$. At small (positive) yaw angles, the power loss by the upstream turbine is less than the power decrease at the downstream turbine. This is due to the upstream turbine deflecting its wake to the side to avoid the downstream turbine. As γ_1 increases past 16 degrees, the increase in downstream turbine power is no longer greater than the loss of power for the upstream wind turbine. The optimal yaw angle that maximizes power production is $\gamma_1^* = 16°$. This increases power production by 5.0% compared to a farm with no yaw-induced wake steering in these wind conditions.

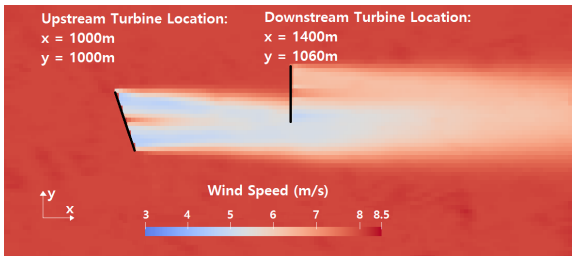


Fig. 4. Turbine locations and hub height 90m wind speed slice for simulated two turbine wind farm simulated in FAST.Farm. The farm-wide low-resolution space spanned 5000m x 2000m x 350m in the x,y, and z directions respectively. Each low-res grid point was separated by 10m and a 2s time step was used. The high-resolution spaces centered around each turbine spanned 160m x 160m x 170m in the x,y, and z directions respectively. Each high-res grid point was separated by 10m and a 0.5s time step was used.

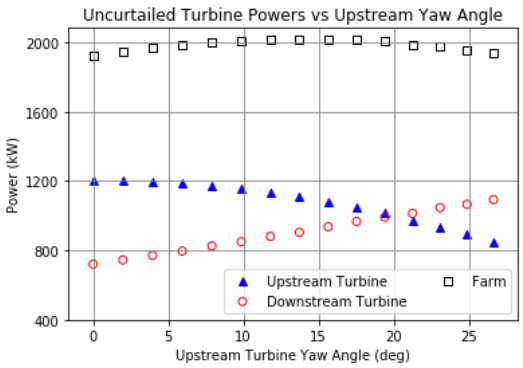


Fig. 5. Uncurtailed baseline exploration shows power outcomes if a yaw exploration experiment could be done in uncurtailed periods. Total farm power increases by 5.0% at an upstream turbine yaw angle of 16°.

D. Problem Statement

The previous section summarized an approach to determine the optimal upstream turbine yaw angle from exploration during uncurtailed periods. This approach has both financial and mechanical risks. If the upstream turbine had

explored to yaw angles that were ineffective at wake steering, the farm would have produced less electricity for an extended period of time, equating to lost revenue. Another risk of the uncurtailed method is that it holds the downstream turbine blade pitch at 0 degrees, increasing the aerodynamic loads that occur as a result of waked wind profiles. These wake profiles may be weighted to one side of the wind turbine during upstream turbine yaw exploration.

The financial and mechanical risks can be mitigated by exploring upstream turbine yaw values during curtailment periods. Curtailment periods require the downstream turbine to pitch its blades to nonzero angles for the purpose of controlling farm power production. Pitched blades capture less wind and therefore be subjected to lower structural loads from upstream wakes [19]. Finally, the goal of wind turbine controllers during curtailment periods is to reduce the efficiency of a wind farm. Hence there would be little to no loss of power production if yaw exploration failed to find a beneficial yaw angle. A key technical challenge with this approach is that it requires an estimate of uncurtailed power output using data obtained during curtailed periods. The following section demonstrates that this prediction can be done with minimal information: a reliable C_p table for the wind turbine in use and accurate measurements of blade pitch, rotor speed, turbine yaw, and turbine power.

III. APPROACH

A. Proposed Exploration Technique

The proposed exploration during curtailment is performed as follows. The turbines initially operate normally by maximizing their own power capture. At some point, a signal is provided by the system operator to curtail the farm power output (in this research: from 1.9 MW down to 1.5 MW). The upstream turbine moves from $\gamma_1 = 0^o$ in the direction predicted to increase power by a computer model, i.e. in the direction of the optimal offset. Throughout the exploration, the downstream turbine continues to hold the total farm power constant using blade pitch control. The power measurements from the upstream turbine are assumed to be available to the downstream turbine controller. The desired power for the downstream turbine $P_{2,desired}$ is the difference between the desired (curtailed) power for the farm minus the power produced by the upstream turbine. The desired downstream generator torque τ_2 is determined from its rotor speed ω_2 as determined by the NREL 5-MW wind turbine's standard torque-speed curve. The desired rotor speed can then be determined by solving the power equation $P_{2,desired} = \tau_2(\omega_2)\omega_2$ for ω_2 where $\tau_2(\omega_2)$ represents generator torque as a function of generator speed. Finally, the downstream turbine tracks this desired rotor speed using blade pitch and a basic PI control.

B. Predicting Uncurtailed Power

The exploration is performed during a curtailed period. The power produced by the farm, upstream, and downstream turbines during curtailment exploration are thus related by:

$$P_{farm,c}(\gamma_1) = P_{1,c}(\gamma_1) + P_{2,c}(\gamma_1) \tag{7}$$

The subscript c describes curtailed operation and, in contrast, u will be used below to denote uncurtailed operation. The downstream turbine uses blade pitch to maintain the curtailed total farm power demand while the upstream turbine performs yaw exploration. Thus $P_{2,c}(\gamma_1)$ compensates for changes in $P_{1,c}(\gamma_1)$ to ensure that $P_{farm,c}(\gamma_1)$ remains near the (constant) desired value.

The objective is to use this exploration during curtailment to determine the optimal upstream yaw offset to maximize total *uncurtailed* power. First note that the upstream turbine is assumed to operate at its peak power coefficient during both curtailed and uncurtailed periods. Hence $P_{1,c}(\gamma_1) = P_{1,u}(\gamma_1)$. However, the downstream turbine is operated differently during the curtailed and uncurtailed periods. The power produced during curtailment must be used to estimate the uncurtailed power. This is done as follows. First, note that the downstream curtailed power is given as:

$$P_{2,c}(\gamma_1) = \frac{1}{2}\rho A U_2(\gamma_1)^3 C_p\left(\beta_c, \frac{R\Omega_c}{U_2(\gamma_1)}\right) \tag{8}$$

The downstream velocity $U_2(\gamma_1)$ depends on the yaw offset of the upstream turbine but it is the same regardless of whether the downstream turbine is in curtailed or uncurtailed operation. This follows because the upstream turbine operates with $C_p(\beta = 0, \lambda = 7.55, \gamma_1)$ in both cases.

The power $P_{2,c}(\gamma_1)$, blade pitch β_c, and rotor speed Ω_c can be measured during curtailed exploration. This leaves the effective wind speed $U_2(\gamma_1)$ as the remaining unknown.[1] This effective wind speed $U_2(\gamma_1)$ can be estimated from Equation 8 using a lookup table for the power coefficient C_P. For this work, a look-up table for the NREL 5MW turbine was constructed using Aerodyn 15 [20] using similar vertical shear that follows the same power law equation shown in equation 6. The C_p lookup-table takes β and λ as inputs and outputs the C_p value. Note, the use of a model-created C_p table signifies this approach is not entirely free of wind turbine modeling. However, the aerodynamic modelling of wind turbine blades is likely more accurate and consistent than wake prediction models, which are made up of aerodynamic models and fluid mechanics models.

Once the incoming effective wind speed has been estimated as $\hat{U}_2$ for the downstream turbine, a predicted power capture for a hypothetical uncurtailed downstream turbine, $\hat{P}_{2,u}(\gamma_1)$, is obtained assuming the maximum power coefficient:

$$\hat{P}_{2,u}(\gamma_1) = \frac{1}{2}\rho A \hat{U}_2(\gamma_1)^3 C_p(\beta = 0, \lambda = 7.55) \tag{9}$$

A model-based estimate for C_p may be used in the equation above as was done to create the C_p lookup-table. Alternatively, Since $\beta = 0$ and $\lambda = 7.55$ occur during all normal turbine operation in region 2 when turbines attempt to maximize their power capture, $C_p(\beta = 0, \lambda = 7.55)$ can be estimated from data collected during normal uncurtailed operation. Specifically, the power produced by the upstream

[1] An anemometer is typically installed on the nacelle. However, this is a single hub height measurement in disrupted flow behind the rotor plane.

turbine during uncurtailed operation at zero yaw offset is:

$$P_{1,u}(\gamma_1 = 0) = \frac{1}{2}\rho A U_1^3 C_p\left(\beta = 0, \frac{\Omega_u R}{U_1}\right) \quad (10)$$

As before, this can be used to estimate the effective wind speed U (the only unknown in equation 10) from measurements of power, blade pitch, and rotor speed. The maximum power coefficient is found simultaneously in this process.

Finally, a prediction for the total power produced by the farm during uncurtailed operation is:

$$\hat{P}_{farm,u}(\gamma_1) = P_{1,u}(\gamma_1) + \hat{P}_{2,u}(\gamma_1) \quad (11)$$

This estimate is formed by exploring to different upstream yaw offsets, γ_1's, during curtailment. The optimal yaw offset for uncurtailed operation is obtained by maximizing this predicted power.

IV. RESULTS AND DISCUSSION

The curtailment exploration experiment was executed as described in Section III. Figure 6 shows the power produced from the upstream turbine, downstream turbine, and their summed farm power at each upstream turbine yaw angle. As expected, the upstream turbine generates less power as it yaws farther from the incoming wind direction. The downstream turbine is controlled in a way that keeps total farm power at 1500 kW.

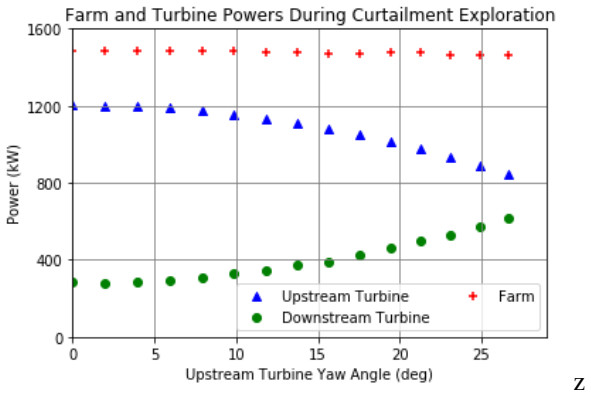


Fig. 6. Power measurements as recorded in FAST.Farm simulation during the exploration experiment as shown in figure 1.

The downstream turbine regulates total farm power by varying its blade pitch angle as shown in Figure 7. Note that the upstream turbine yaw angle corresponding to the maximum downstream blade pitch is not the yaw angle that will optimize power under normal uncurtailed farm operation. The maximum blade pitch signifies the largest change in how much energy is captured from the wind comparing curtailed to uncurtailed states. However, as the upstream turbine yaw angle increases, the power in the wind also increases. Here, the upstream yaw angle of 10 degrees forces the maximum downstream blade pitch, but the maximum farm power output is predicted to occur at 14 degrees. This small difference between predicted and actual optimal yaw angle is likely due to the C_p table not being as accurate at predicting the efficiency of a partially waked wind turbine. Small inconsistencies between simulations may also be present. A C_p table for varying incoming wind profiles that exhibit high horizontal shear due to wakes may help to increase prediction accuracy. A smaller step between yaw angles or a nonlinear optimization gradient descent may allow for greater precision by simply taking more data points near the optimal yaw angle.

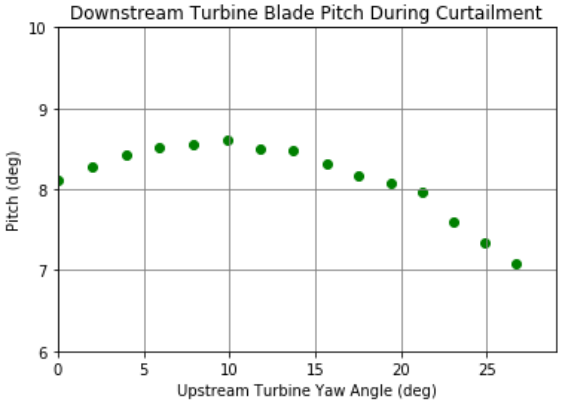


Fig. 7. Downstream turbine blade pitch changes to regulate farm power to a constant 1.5MW while the upstream turbine explores to different yaw states.

Note the maximum blade pitch for the downstream turbine does not correspond to the optimal yaw angle. Here, the maximum downstream blade pitch angle occurs at a 10 degree upstream yaw angle.

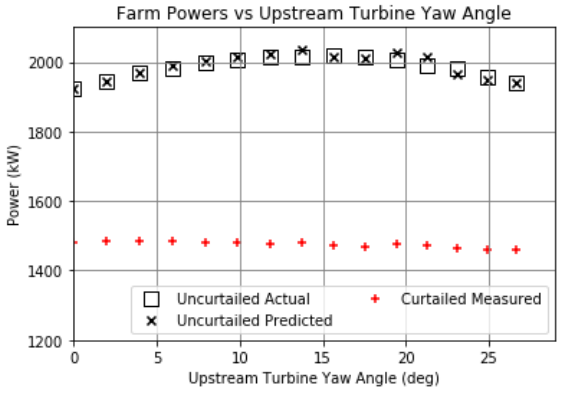


Fig. 8. C_p ratios predict total farm power from a curtailed farm power, predicting an optimal upstream turbine yaw angle of 14 degrees.

Figure 8 shows the power produced by the wind farm during exploration at various upstream yaw offsets. The plus signs are the measured power during curtailment and the black x's are the predicted uncurtailed power. The actual uncurtailed power (shown as black squares) would be unknown except the far left black square in Figure 8 is assumed to be a known quantity since it represents an uncurtailed farm operating with a 0 degree upstream turbine yaw angle and a 0 degree downstream turbine pitch (standard turbine operation). Equation 10 gives an uncurtailed C_p = 0.45, a small but significant adjustment to the theoretical

maximum efficiency for the NREL 5 MW turbine given as 0.48. Equation 9 is then solved for all other yaw angles to predict downstream turbine power if it was not curtailed using blade pitch control.

The predicted downstream power is then added to the upstream turbine measured power output to produce a predicted farm power as a function of upstream yaw angle, shown as the x's in Figure 8.

V. CONCLUSION AND FUTURE WORK

Grid curtailment periods offer an opportunities to explore possible wake-steering states to find the optimal yaw angle for maximum power production with minimal mechanical and financial risks. A simple rotor effective wind speed can calculate C_p ratios used to predict uncurtailed power from measured curtailed power. An optimal upstream turbine yaw angle of 14 degrees was predicted from the curtailment exploration with the true optimal angle was 16 degrees. Accuracy may increase if different C_p tables are utilized for each magnitude of horizontal wind shear due to partial waking of the downstream turbine. This method requires only blade pitch angle, rotor speed, and yaw angle measurements on the wind turbines.

The simplicity of this method is made possible by assuming a known C_p table and highly consistent incoming wind. Future work will analyze how more realistic incoming wind profiles may be accounted for. The disconnect between a model-based C_p table and realistic turbine operation will be bridged. This includes generating a C_p table from simulated turbine operation that more resembles real-world operation. This includes sensor noise, a limited number of sensors, and more realistic wind profiles. The utilization of this C_p table for curtailment exploration should be applied in more realistic wind farm conditions and operation as well in future research for validation. Using wind turbine blade bending stress methods [21] may prove useful for C_p table utilization in future work.

Future work may also look to use similar wake-steering exploration techniques during other times when wind turbines are controlled to work less efficiently such as in wind regions 1 and 3. Larger wind farms will be simulated and analyzed to show how these exploration techniques may be executed with deep-array effects. Loading will be analyzed to assess how different yaw angles impact fatigue stresses on both the yawing turbine and the downstream turbine. The change in loads for the downstream turbine as it pitches its blades during curtailment exploration is also of interest for future work.

VI. ACKNOWLEDGMENTS

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